Answers to some recent conjectures and questions in classical analysis, with computer assistance

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Abstract

We answer, or make progress on, several questions and conjectures posed in recent arXiv papers in classical analysis. We prove two conjectures of Fan on Riesz capacity of three-point sets on the circle and of vertex sets of regular odd polygons. We show that a conjecture of Xu and Zhao on the sign thresholds of higher-order Gregory coefficients is false and compute these thresholds rigorously for orders up to 10 (taking an integral identity of Xu and Zhao as given). We prove by computer-assisted ball arithmetic that the reciprocal of the diagonal Padé denominator of e^z has positive Taylor coefficients for every odd degree $3 \le m \le 361$, a case left open in the literature, and describe numerically where positivity first fails for fixed m. We also give explicit values for series of products of consecutive harmonic numbers, short answers to questions of Finch on Abel's equation and iterates, two of three open problems of Wiśniewska on Gurland's ratio, and numerical values of two completely monotonic degrees. Theorems, computer-assisted proofs and numerical results are clearly separated. The work was carried out with extensive use of an AI system (see the statement on the use of AI).

1 Introduction

This paper collects answers to a number of specific questions and conjectures that appeared in recent arXiv papers (2021–2026) in classical analysis: inequalities and sharp constants, series and sums, special functions, iteration and asymptotics. The questions were chosen from the literature by an AI system, by the rule that someone had written the question down (as a conjecture, an open problem, or a request for a value) and that it could be attacked by calculus together with computer algebra, interval arithmetic or exact computation. For each question we state the original wording and its source, we state our result, and we give the proof or, for numerical results, the method and the precision. We distinguish three kinds of statements throughout.

- **Theorems**: complete proofs in the text (Section 2; Propositions 6.1 and 7.1, which are short and elementary).
- **Computer-assisted theorems**: statements whose proof consists of a written argument plus a finite computation in exact or interval arithmetic that we describe precisely (Theorems 3.2(ii) and 4.2, and the exact reductions in Section 5).

*Directing author. The mathematical work was carried out by an AI system, Claude (Anthropic), via Claude Code; see the statement on the use of AI at the end of Section 1.

- **Numerical results**: values or laws obtained by high-precision computation and checked by independent code, but not proved (for example the tables in Sections 6 and 8 and the law for $k^*(m)$ in Section 4.4). These are never presented as theorems.

Summary of results.

1. (Section 2) We prove two conjectures of Fan [1] on Riesz capacity of three-point sets on a circle and of the vertex sets of regular odd polygons (Theorem 2.3).

2. (Section 3) We show that a conjecture of Xu and Zhao [2] on the sign thresholds $\mathrm{Gr}(\ell)$ of higher-order Gregory coefficients is false, compute $\mathrm{Gr}(\ell)$ rigorously for $\ell \le 10$ (and numerically up to 14), and describe the observed growth.

3. (Section 4) For the denominators $q_{m,m}$ of diagonal Padé approximants to e^z we prove, by computer assistance, that $1/q_{m,m}$ has positive Taylor coefficients for every odd $3 \le m \le 361$, a case left open in [4] of a conjecture of Fasi and Higham [3], and we describe numerically where positivity first fails for fixed m.

4. (Section 5) For the series $S_q = \sum_{k\ge1} H_k H_{k+1} \cdots H_{k+q}/(k(k+1)\cdots(k+q))$ studied in [5] we describe an algorithm that reduces S_q exactly (in rational arithmetic, executed for $q \le 9$) to multiple zeta values, and we give explicit closed forms in zeta values for $q \le 8$; the last step (identifying the multiple zeta values) is a numerical identification, not a proof.

5. (Section 6) We give a derivation of Finch's conjectured values of the constant between two solutions of Abel's equation (an easy consequence of classical Fatou-coordinate asymptotics), an exact series and the value of a second derivative asked for in [9], and half-iterate values asked for in [10].

6. (Section 7) We answer two of the three open problems of Wiśniewska [13] on Gurland's ratio (by an elementary argument) and give numerical information on the third.

7. (Section 8) We give numerical values of two completely monotonic degrees whose exact values are said to be open in [14].

Results that turned out to be known. We also found proofs of three statements that, as our literature search showed, had already been settled, and which we therefore do not claim: Conjecture 7.1 of Habib and Hildebrand [15] (proved by the same authors a few days later in [16]); Conjecture 4.5 of Chavan, Kobayashi and Layja [17] on the multiple t-values $t(\{2\}^N, 1)$ (proved by Charlton [18]); and the two conjectured identities between Abel functions in [11] (explained by the author himself in the corrigendum of that paper). A numerical refutation of a Mahler-measure conjecture and its explanation, which we also reproduced, are in [19]. We mention them to give an accurate account of what the search found.

Limits of the literature search. Novelty statements are based on web and arXiv searches carried out in November 2026; citation-count tools were mostly unavailable, so a later or less visible reference to a result may have been missed. "Not found" means only that. Several papers cited here appeared in 2026; they are cited by arXiv number and were read in their arXiv versions.

Code availability. All code, logs and verification scripts are in the repository https://github.com/JacobGoodchild/findformula3, under `results/`; the file `VERIFICATION.md` lists, for each result, what was checked and by which method.

Statement on the use of AI. This work was carried out with extensive use of an AI system, Claude (Anthropic), via the Claude Code software, under the direction of Jacob Goodchild. The AI system chose the problems from the recent literature, carried out the derivations, proofs and computations, searched and checked the literature, and wrote the manuscript. Proofs were checked by independent AI referee passes and by computer algebra and interval arithmetic, but have not been checked by a human expert or peer reviewed. Novelty assessments are based on literature searches and may be incomplete. Jacob Goodchild directed the work and is responsible for its publication.

Verification status. The proofs in Sections 2, 6.1 and 7 are written out in full and rely, besides standard facts, on the three results quoted from [1] in Section 2. The computer-assisted statements (Theorems 3.2(ii), 4.2, 5.1) rest on ball-arithmetic or exact-rational computations that we describe; the closed forms in Section 5 are numerical identifications. The values in Sections 6 (the second derivative and the half-iterates) and 8, and the laws for $k^*(m)$ and for the growth of $\mathrm{Gr}(\ell)$, are numerical, computed twice by independent programs for the key values, and *not* interval-certified.

2 Riesz energy of three points on a circle

2.1 The question

Let $K \subset \mathbb{R}^2$ be compact and $r > 0$. Write

$$U_r(K) \;=\; \max_{\mu} \iint |x-y|^r \,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y),$$

the maximum over Borel probability measures μ on K (a maximiser is called an *equilibrium measure*). In the notation of Fan [1], $U_r(K) = V_{-r}(K)$ and the Riesz capacity is $\mathrm{Cap}_{-r}(K) = U_r(K)^{1/r}$, so maximising capacity and maximising U_r are the same problem. Fan studies the vertex set P_N of the regular N-gon inscribed in the unit circle and states two conjectures, which we quote.

Conjecture 2.1 (Fan [1], Conjecture 5.9)**.** Suppose $r \geq 2$. Let A, B, C be three points on the unit circle. Fix B and C such that they are symmetric with respect to the x-axis, having polar angles φ and $-\varphi$, respectively, where $\varphi \in (\pi/2, 2\pi/3]$. Move A along the circle with polar angle $\psi \in [0, 2\pi - 3\varphi]$, so that BC remains the longest side of the triangle $\triangle ABC$. Then the capacity of the three-point set $\{A, B, C\}$ is maximized when $\psi = 2\pi - 3\varphi$, i.e., when $AC = BC$.

Conjecture 2.2 (Fan [1], Conjecture 5.6)**.** Let $r > 2$ and let P_N be the vertex set of a regular N-gon inscribed in the unit circle ($N \geq 3$) with N odd. Let $\theta_N = \frac{(N-1)\pi}{2N}$. Then the energy equals $U_r(P_N) = \dfrac{2^{r-1}\sin^r \theta_N}{1 - 2^{r-2}\cos^r \theta_N}$, and an equilibrium measure is supported on the three points 1, $e^{2\theta_N i}$, $e^{2(\pi-\theta_N)i}$ (up to rotation), with masses $\dfrac{2b^r - a^r}{4b^r - a^r}$, $\dfrac{b^r}{4b^r - a^r}$, $\dfrac{b^r}{4b^r - a^r}$ respectively, where $a = 2\sin(\pi/N)$ and $b = 2\sin\theta_N$.

Fan proves the special case of Conjecture 2.1 in which A is replaced by the symmetric configuration (her Lemma 5.8, restated below), shows that Conjecture 2.1 together with that lemma implies Conjecture 2.2, and describes her attempt to prove Conjecture 2.1 as unsuccessful. We prove both conjectures. The proof is a calculus argument; its only external inputs are three results quoted in [1], which we use as stated there.

Theorem 2.3. *Conjecture 2.1 and Conjecture 2.2 are true. More precisely, in the setting of Conjecture 2.1, the energy $U_r(\{A,B,C\})$ is a nondecreasing function of $\psi \in [0, 2\pi - 3\varphi]$.*

Notation. In Conjecture 2.2 the symbols a, b are Fan's; in the proof below they correspond to β and ℓ. Elsewhere a, b, c denote side lengths of a triangle.

2.2 Inputs taken from the literature

(I1) *Energy of three points* (Clark–Laugesen [20], Theorem 8, quoted as Lemma 5.7 in [1]). Let $T \subset \mathbb{R}^2$ have three points with mutual distances a, b, c, $0 < a, b \le c$, and let $r > 0$. If $a^r + b^r \le c^r$ then $U_r(T) = c^r/2$; otherwise

$$U_r(T) = \frac{2a^r b^r c^r}{4a^r b^r - (a^r + b^r - c^r)^2}.$$

(I2) *Support* (Björck [21], Theorem 12, quoted in [1] before Lemma 5.2). If $r > 2$, every compact $K \subset \mathbb{R}^2$ has an equilibrium measure supported on at most three points, which are extreme points of the convex hull of K. In particular $U_r(P_N) = \max\{U_r(T) : T \subset P_N,\ |T| \le 3\}$.

(I3) *Isosceles triples* (Fan, Lemma 5.8). Let $r \ge 2$, $A = (1, 0)$, $B = e^{i\alpha}$, $C = e^{-i\alpha}$. Then $U_r(\{A, B, C\})$ is nondecreasing in $\alpha \in [2\pi/3, \pi)$.

We did not open the papers of Björck and Clark–Laugesen; (I1) and (I2) are used exactly as they are stated in [1].

2.3 Notation and a formula for the derivative

Throughout, $r \ge 2$. For a triple with sides $a, b \le c$ put $x = a^r$, $y = b^r$, $z = c^r$ and $s = x + y - z$. If $s \le 0$ we are on the *two-point branch*, $U = z/2$; if $s > 0$ we are on the *three-point branch*, and then, since $z \ge \max(x, y)$ gives $s \le \min(x, y)$,

$$D := 4xy - s^2 \ \ge\ 4xy - \min(x,y)^2 \ > 0, \qquad U = \frac{2xyz}{D}, \qquad U - \frac{z}{2} = \frac{z\,s^2}{2D} > 0.$$

At $s = 0$ both expressions equal $z/2$, so U is a continuous function of (x, y, z).

Fix $\varphi \in (\pi/2, 2\pi/3]$ and let $B = e^{i\varphi}$, $C = e^{-i\varphi}$, $A = e^{i\psi}$ with $0 \le \psi \le 2\pi - 3\varphi$. Put

$$\lambda = \frac{\varphi - \psi}{2}, \qquad \eta = \frac{\varphi + \psi}{2}, \qquad \text{so} \qquad 0 \le \lambda \le \eta \le \pi - \varphi < \frac{\pi}{2} < \varphi.$$

(The inequality $\eta \le \pi - \varphi$ is $\psi \le 2\pi - 3\varphi$; $\lambda > 0$ for $\psi < \varphi$, which holds because $2\pi - 3\varphi < \varphi$ when $\varphi > \pi/2$.) The chord lengths are

$$a := |AB| = 2\sin\lambda, \qquad b := |AC| = 2\sin\eta, \qquad c := |BC| = 2\sin\varphi.$$

Since sin is increasing on $[0,\pi/2]$ and $\sin\varphi = \sin(\pi-\varphi)$, we have $a \le b \le c$, with $b = c$ exactly at $\psi = 2\pi - 3\varphi$ and $a = b$ exactly at $\psi = 0$. So BC is the longest side, as in Conjecture 2.1. As ψ increases, a decreases and b increases, c is constant. Write $F(\psi) = U_r(\{A,B,C\})$; F is continuous.

Lemma 2.4 (derivative)**.** *On a ψ-interval where $s > 0$ and $0 < \psi < 2\pi - 3\varphi$ (so $0 < \lambda < \eta$ and $b < c$),*

$$\frac{\mathrm{d}\log F}{\mathrm{d}\psi} = \frac{s\,r}{2D}\Big[(z+y-x)\cot\eta - (z+x-y)\cot\lambda\Big].$$

Proof. From $\log U = \log 2 + \log x + \log y + \log z - \log D$ and $D = 4xy - s^2$, $s = x+y-z$,

$$\frac{\partial \log U}{\partial x} = \frac{1}{x} - \frac{4y-2s}{D} = \frac{D - x(4y-2s)}{xD} = \frac{s(2x-s)}{xD} = \frac{s\,(z+x-y)}{xD},$$

and symmetrically $\partial_y \log U = s(z+y-x)/(yD)$. Now $\mathrm{d}\lambda/\mathrm{d}\psi = -\frac12$, $\mathrm{d}\eta/\mathrm{d}\psi = \frac12$, so $\mathrm{d}\log x/\mathrm{d}\psi = -\frac{r}{2}\cot\lambda$ and $\mathrm{d}\log y/\mathrm{d}\psi = \frac{r}{2}\cot\eta$, while z does not depend on ψ. The chain rule gives the formula. □

2.4 Monotonicity of F

Lemma 2.5 (an elementary inequality)**.** *Let $0 < p < q < 1$, let $r \ge 2$ and suppose $p^r + q^r > 1$. Then $q^r - p^r \;\ge\; q^2 - p^2$.*

Proof. Put $G(\rho) = q^\rho - p^\rho - (q^2 - p^2)$ for real $\rho \ge 2$, so $G(2) = 0$ and $G'(\rho) = q^\rho \ln q - p^\rho \ln p$. Since $\ln p, \ln q < 0$, $G'(\rho) \ge 0$ is equivalent to $(q/p)^\rho \le \ln p / \ln q$. The left side is strictly increasing in ρ (because $q > p$), so the set $\{\rho \ge 2 : G'(\rho) \ge 0\}$ is an initial segment of $[2,\infty)$ (possibly empty): G is nondecreasing on it and strictly decreasing afterwards. In either case, on every interval $[2,\rho_1]$ we have $G \ge \min\{G(2), G(\rho_1)\}$.

The function $\rho \mapsto p^\rho + q^\rho$ is strictly decreasing and tends to 0 as $\rho \to \infty$; let ρ_1 be the root of $p^{\rho_1} + q^{\rho_1} = 1$. Because $p^r + q^r > 1$ we have $2 \le r < \rho_1$, so it suffices to prove $G(\rho_1) \ge 0$.

Put $w = q^{\rho_1}$. Then $p^{\rho_1} = 1 - w$, $w \in (\frac12, 1)$ (as $q > p$), and with $\sigma = 2/\rho_1 \in (0,1]$ we have $q^2 = w^\sigma$, $p^2 = (1-w)^\sigma$. Thus

$$G(\rho_1) = (2w-1) - \big(w^\sigma - (1-w)^\sigma\big) = \chi(1) - \chi(\sigma), \qquad \chi(\sigma) := w^\sigma - (1-w)^\sigma,$$

and it suffices to show that χ is nondecreasing on $(0,1]$. We have $\chi'(\sigma) = (1-w)^\sigma |\ln(1-w)| - w^\sigma |\ln w|$, so

$$\frac{\chi'(\sigma)}{w^\sigma |\ln w|} = \Big(\frac{1-w}{w}\Big)^\sigma \frac{|\ln(1-w)|}{|\ln w|} - 1 \;\ge\; \frac{(1-w)|\ln(1-w)|}{w|\ln w|} - 1 = \frac{k(1-w)}{k(w)} - 1,$$

where $k(u) = -u\ln u$ and we used $\frac{1-w}{w} < 1$, $\sigma \le 1$. It remains to see $k(1-w) \ge k(w)$ for $w \in [\frac12, 1)$. Put $u = 1 - w \in (0, \frac12]$ and $E(u) = k(u) - k(1-u)$. Then E is continuous on $[0,\frac12]$ with $E(0) = E(\frac12) = 0$, and $E''(u) = -\frac{1}{u} + \frac{1}{1-u} = \frac{2u-1}{u(1-u)} < 0$ on $(0,\frac12)$, so E is concave and $E \ge 0$. Hence $\chi' \ge 0$. □

Proposition 2.6. *F is nondecreasing on $[0, 2\pi - 3\varphi]$.*

Proof. On the open set where $s > 0$ and $0 < \psi < 2\pi - 3\varphi$ we use Lemma 2.4. Here $s > 0$, $D > 0$, $\cot\eta > 0$ (as $\eta < \pi/2$), $\cot\lambda > 0$, and $z + x - y \geq x > 0$ (as $y \leq z$). So the sign of $\mathrm{d}\log F/\mathrm{d}\psi$ is the sign of

$$R_r - \frac{\cot\lambda}{\cot\eta}, \qquad R_r := \frac{z+y-x}{z+x-y}.$$

Put $p = a/c$, $q = b/c$ (so $0 < p < q < 1$) and $t_r = q^r - p^r \in (-1, 1)$. Then $R_r = \dfrac{1+t_r}{1-t_r}$, which is an increasing function of t_r. For $r = 2$ we compute with $\varphi = \lambda + \eta$ and $\sin^2\eta - \sin^2\lambda = \sin(\eta-\lambda)\sin(\eta+\lambda) = \sin\psi\sin\varphi$:

$$R_2 = \frac{\sin^2\varphi + \sin\varphi\sin\psi}{\sin^2\varphi - \sin\varphi\sin\psi} = \frac{\sin\varphi + \sin\psi}{\sin\varphi - \sin\psi} = \frac{2\sin\eta\cos\lambda}{2\cos\eta\sin\lambda} = \frac{\cot\lambda}{\cot\eta},$$

where $\sin\varphi + \sin\psi = 2\sin\eta\cos\lambda$ and $\sin\varphi - \sin\psi = 2\cos\eta\sin\lambda$. Hence the sign of $\mathrm{d}\log F/\mathrm{d}\psi$ is the sign of $R_r - R_2$, i.e. of $t_r - t_2$, i.e. of $q^r - p^r - (q^2 - p^2)$.

We check the hypotheses of Lemma 2.5: $0 < p < q < 1$ was noted above, and $p^r + q^r > 1$ is exactly $s > 0$. By Lemma 2.5, $t_r \geq t_2$, hence $\mathrm{d}\log F/\mathrm{d}\psi \geq 0$ wherever $s > 0$ in the open interval.

It remains to treat the branches. Suppose $s(\psi_1) > 0$ for some $\psi_1 \in [0, 2\pi - 3\varphi)$, and let $\psi_2 = \sup\{\psi' \in (\psi_1, 2\pi - 3\varphi] : s > 0 \text{ on } [\psi_1, \psi')\}$. On $[\psi_1, \psi_2)$ the function F is continuous and nondecreasing (it is differentiable with nonnegative derivative on the open set where $s > 0$ and $0 < \psi < 2\pi - 3\varphi$), so $F(\psi_2) \geq F(\psi_1) > z/2$ by continuity. If $\psi_2 < 2\pi - 3\varphi$ then $s(\psi_2) = 0$ and $F(\psi_2) = z/2$, a contradiction; hence $\psi_2 = 2\pi - 3\varphi$, i.e. once $s > 0$ it stays positive up to the endpoint, and F is nondecreasing on $[\psi_1, 2\pi - 3\varphi]$. On the set where $s \leq 0$, $F \equiv z/2$ is constant, and $s > 0$ at ψ implies $s > 0$ later, so this set is an initial interval $[0, \psi_*]$ (possibly empty) on which $F = z/2$. Since F is continuous and $F \geq z/2$ always, F is nondecreasing on $[0, 2\pi - 3\varphi]$. The degenerate case $\varphi = 2\pi/3$ (where $2\pi - 3\varphi = 0$) is trivial. □

Proof of Conjecture 2.1. By Proposition 2.6, $F(\psi) \leq F(2\pi - 3\varphi)$, i.e. the energy, and hence the capacity $U_r^{1/r}$, is maximal at $\psi = 2\pi - 3\varphi$, where $b = c$ (that is $AC = BC$). □

Remark 2.7. The inequality of Lemma 2.5 gives only $\geq$, so we claim monotonicity (not strict monotonicity) of F, which is all that Conjecture 2.1 asserts. At $r = 2$, $t_2 - t_2 = 0$ and F is constant, as in Theorem 5.4 of [1].

2.5 Proof of Conjecture 2.2

Let $N \geq 3$ be odd, $r > 2$, $\theta = \theta_N = \frac{\pi}{2} - \frac{\pi}{2N}$, and let $v_k = e^{2\pi i k/N}$. Put $\ell = 2\sin\theta$ (the longest chord of P_N) and $\beta = 2\sin(\pi/N)$. Since $\pi/N = \pi - 2\theta$ we have $\beta = 2\sin 2\theta = 4\sin\theta\cos\theta$.

The conjectured triple. Let $T_N = \{1, e^{i(\pi-\pi/N)}, e^{i(\pi+\pi/N)}\}$; this is the triple of Conjecture 2.2 because $2\theta_N = \pi - \pi/N$ and $2(\pi - \theta_N) = \pi + \pi/N$. For odd N these are vertices of P_N ($\frac{N\mp1}{2}$ are integers). T_N is isosceles with two sides ℓ and base $\beta \leq \ell$ (this is $\sin\frac{\pi}{2N} \leq \frac{1}{2}$, true for $N \geq 3$). Applying (I1) with $c = \ell$ and the two other sides ℓ, β (so $\ell^r + \beta^r > \ell^r$: three-point branch),

$$U_r(T_N) = \frac{2\ell^{2r}\beta^r}{4\ell^r\beta^r - \beta^{2r}} = \frac{2\ell^{2r}}{4\ell^r - \beta^r} = \frac{2^{r-1}\sin^r\theta}{1 - 2^{r-2}\cos^r\theta},$$

where the last equality uses $\ell = 2\sin\theta$, $\beta = 4\sin\theta\cos\theta$. This is the energy claimed in Conjecture 2.2. Moreover let μ_N be the measure on T_N with the masses of Conjecture 2.2 (the

apex 1 carries $\frac{2\ell^r-\beta^r}{4\ell^r-\beta^r}$, and the other two points carry $\frac{\ell^r}{4\ell^r-\beta^r}$ each; here $a=\beta$, $b=\ell$ in the notation of the conjecture). These masses are positive ($2\ell^r>\beta^r$) and sum to 1, and a direct computation gives

$$\iint |x-y|^r \mathrm{d}\mu_N \mathrm{d}\mu_N = 2\Big[2\,m_1 m_2 \ell^r + m_2^2 \beta^r\Big] = \frac{2\ell^{2r}\,(4\ell^r-\beta^r)}{(4\ell^r-\beta^r)^2} = U_r(T_N).$$

Upper bound. By (I2) it suffices to show $U_r(T) \le U_r(T_N)$ for every $T \subset P_N$ with at most three points. Single points give $U_r = 0$, and for a pair at distance d one has $U_r = \max_{0\le p\le 1} 2p(1-p)d^r = d^r/2$, so pairs give $U_r = d^r/2 \le \ell^r/2 \le U_r(T_N)$ (where $d \le \ell$ is the distance, and $U_r(T_N) \ge \ell^r/2$ because $U \ge z/2$ in (I1)). Let $T=\{A,B,C\} \subset P_N$ be a triple with longest side BC.

- If $a^r + b^r \le c^r$ (two-point branch), $U_r(T) = c^r/2 \le \ell^r/2 \le U_r(T_N)$.
- If $a^r+b^r > c^r$, then, since $(a/c)^r \le (a/c)^2$ and $(b/c)^r \le (b/c)^2$ for $r>2$, we get $a^2+b^2>c^2$, so all angles of the triangle are acute and T is an acute inscribed triangle: its three arcs are all smaller than π. Rotate so that the midpoint of the arc BC not containing A is $e^{i\pi}$; if that arc has length $2\pi-2\varphi$ (which is $<\pi$, so $\varphi>\pi/2$) then $B,C = e^{\pm i\varphi}$ (after naming), and $A=e^{i\psi}$ lies on the complementary arc, $|\psi|<\varphi$; after a reflection in the x-axis, $\psi \in [0,\varphi)$; the arcs AB, AC have lengths $\varphi-\psi$, $\varphi+\psi$; BC longest means $2\pi-2\varphi \ge \varphi+\psi$, i.e. $\psi \le 2\pi-3\varphi$; and since the three arcs sum to 2π the longest is at least $2\pi/3$, so $\varphi \le 2\pi/3$. Thus T is exactly in the configuration of Theorem 2.3 and, by Proposition 2.6, $U_r(T) \le U_r(T')$ where T' is the configuration with $\psi = 2\pi-3\varphi$, i.e. $|AC| = |BC| =: L = 2\sin\varphi$. An isosceles triangle inscribed in the unit circle with legs L is unique up to congruence (its apex is a point of the circle and the other two vertices are the two points of the circle at distance L from it), hence T' is congruent to $\{(1,0), e^{i\alpha}, e^{-i\alpha}\}$ with $2\sin(\alpha/2) = L$, i.e. $\alpha = 2\pi-2\varphi \in [2\pi/3, \pi)$ (as $\varphi \in (\pi/2, 2\pi/3]$). As $L=|BC|$ is a chord of P_N, $L \le \ell = 2\sin\theta$, so $\alpha/2 \le \theta$ (both lie in $(0,\pi/2)$) and $\alpha \le 2\theta = \pi - \pi/N$, where $2\theta \ge 2\pi/3$ because $N \ge 3$. By (I3), $U_r(T') \le U_r(\{(1,0), e^{i(\pi-\pi/N)}, e^{-i(\pi-\pi/N)}\}) = U_r(T_N)$.

Hence $U_r(P_N) \le U_r(T_N) = \iint |x-y|^r \mathrm{d}\mu_N \mathrm{d}\mu_N \le U_r(P_N)$, so $U_r(P_N) = U_r(T_N)$ and μ_N is an equilibrium measure of P_N, supported on the three points of Conjecture 2.2. This proves Conjecture 2.2. □

Remark 2.8. We prove existence of an equilibrium measure with the stated support and masses; we do not claim uniqueness up to rotation (Fan notes the same limitation of her numerical evidence). When $3 \mid N$ the equilateral triangle of vertices is an admissible acute triple covered by the case analysis above.

2.6 Checks

The derivative formula, Lemma 2.5, and the assembly of $U_r(T_N)$ were checked with computer algebra and high-precision numerics, and the conclusion of Conjecture 2.2 was compared with a brute-force maximisation over all vertex triples for odd $N \le 31$ and several values of $r>2$ (see the repository, `results/riesz_triangle`). These checks are not part of the proof.

3 Sign thresholds of higher-order Gregory coefficients

3.1 The question

Let $G_n^{(\ell)}$ be the Taylor coefficients of $(x/\ln(1+x))^\ell = \sum_{n\ge 0} G_n^{(\ell)} x^n$ (higher-order Gregory coefficients; $\ell = 1$ gives the classical Gregory coefficients). Xu and Zhao [2] prove (their Theorem 5.1) that for each $\ell \ge 1$ there is a positive integer $\mathrm{Gr}(\ell)$ such that $(-1)^{n-1} G_n^{(\ell)} > 0$ for all $n \ge \mathrm{Gr}(\ell)$. Their wording only asserts the existence of *some* integer; they then quote "exact values" $\mathrm{Gr}(3) = 11$, $\mathrm{Gr}(4) = 36$, $\mathrm{Gr}(5) = 113$, $\mathrm{Gr}(6) = 346$, $\mathrm{Gr}(7) = 1030$, so we read $\mathrm{Gr}(\ell)$ as the *least* such integer, i.e. $1+$the largest $n \ge 1$ with $(-1)^{n-1} G_n^{(\ell)} \le 0$. (A few small-$n$ coefficients vanish, e.g. $G_{\ell+1}^{(\ell)} = 0$ for even ℓ and $G_\ell^{(\ell)} = 0$ for odd $\ell \ge 3$; they lie far below the last bad n.) They ask:

Conjecture 3.1 (Xu–Zhao [2], Conjecture 6.1)**.** For all $\ell \ge 5$ we have $0.47 \cdot 3^\ell < \mathrm{Gr}(\ell) < 0.477 \cdot 3^\ell$.

In the same passage the authors remark that the largest real root $r(\ell)$ of the function appearing in their proof "seems to increase by roughly tripling as ℓ increases, similarly to $\{\mathrm{Gr}(\ell)\}$", and ask whether there is a theoretical reason.

3.2 Results

Theorem 3.2. *Conjecture 3.1 is false. Indeed:*

(i) $\mathrm{Gr}(5) = 113 < 114.21 = 0.47 \cdot 3^5$. *This follows from the value* 113 *stated in [2]; we also reproduce it.*

(ii) (computer-assisted, rigorous) $\mathrm{Gr}(\ell)$ *equals* $11, 36, 113, 346, 1030, 2995, 8580, 24303$ *for* $\ell = 3, \dots, 10$. *Thus* $\mathrm{Gr}(\ell) < 0.47 \cdot 3^\ell$ *also for* $\ell = 8, 9, 10$ *(*3083.7*,* 9251.0*,* 27753.0 *against* 2995*,* 8580*,* 24303*). The values for* $\ell = 3, \dots, 7$ *agree with [2]; those for* $\ell = 8, 9, 10$ *are new.*

Proof of (i). A one-line comparison of the stated value with the stated lower bound.
Proof of (ii) (computer-assisted). The argument has two parts, as in the proof of Theorem 5.1 of [2].

Finite range. The coefficients $G_n^{(\ell)}$, $n \le N_{\max}$, were computed as ball-valued power series (Arb through python-flint, via $\exp(-\ell \log(\log(1+x)/x))$) with every sign certified; the working precision was raised until all coefficients with $n \ge 50$ had a definite sign (128 bits sufficed in the independent run below). This gives the largest $n \le N_{\max}$ with the wrong sign.

Tail. For $n > \ell + 3$, [2] writes $(-1)^{n-1} G_n^{(\ell)} = \int_1^\infty h(z) f_n(z)\,\mathrm{d}z$ (their identity (6)) with explicit h, f_n, and, by cutting the integral at the largest real root $r = r(\ell)$ of h and bounding the pieces, shows that $(-1)^{n-1} G_n^{(\ell)} > 0$ for all $n \ge n_2(\ell)$, where $n_2(\ell)$ is the least n for which an explicit inequality involving integrals over $[1, r]$ and $[0, 1/r]$ holds (an inequality which is monotone in n). Evaluating that inequality with rigorous interval integration (Arb), with certified isolation of the $\ell - 1$ real roots of the polynomial H_ℓ of [2], gives $n_2(\ell) = 15, 52, 168, 519, 1548, 4514, 12944, 36685$ for $\ell = 3, \dots, 10$, with positive certified margins; for $\ell = 4, \dots, 10$ these are the values of Table 2 of [2] ($\ell = 3$: 15 instead of their 16). All are below

the ranges scanned ($N_{\max} \geq 6000$ for $\ell \leq 8$ and ≥ 45000 for $\ell = 9, 10$). The values of $\mathrm{Gr}(\ell)$ then follow. □

We stress what this proof takes from [2]: the integral identity (6), the structure of the inequality and its monotonicity in n (we re-checked (6) numerically against exact coefficients and the root counts by certified isolation, but did not re-prove them). The sign scan and the tail evaluation were carried out twice, by independent implementations (the second written without access to the first), with the same values of $\mathrm{Gr}(\ell)$ and, for $\ell = 5, \ldots, 10$, the same thresholds $n_2(\ell)$ (our first, non-rigorous implementation gave $18, 53$ instead of $15/16$, 52 for $\ell = 3, 4$; this does not affect $\mathrm{Gr}(3), \mathrm{Gr}(4)$).

Remark 3.3. Table 1 of [2] lists $r(9) \approx 5065$; with $r(\ell) = \exp(\pi \cot(\pi/\ell))$ (see below) one gets $r(9) = 5605.2$, so we believe 5065 is a transposition of digits (their $n_2(9)$ agrees with ours).

3.3 More values and the growth of $\mathrm{Gr}(\ell)$

The same scan, with the tail inequality evaluated numerically (not rigorously, and without certified margins) gives $\mathrm{Gr}(11) = 68283$, $\mathrm{Gr}(12) = 190698$, $\mathrm{Gr}(13) = 530152$, $\mathrm{Gr}(14) = 1468691$; an independent sign scan (to $N = 1.2 \cdot 10^5$) reproduced $\mathrm{Gr}(11)$. Table 1 lists the data together with $r(\ell) = \exp(\pi \cot(\pi/\ell))$.

Table 1: $\mathrm{Gr}(\ell)$ for $3 \leq \ell \leq 14$ (rigorous for $\ell \leq 10$; numerical for $\ell = 11, \ldots, 14$). $r(\ell) = \exp(\pi \cot(\pi/\ell))$.

ℓ	$\mathrm{Gr}(\ell)$	$\mathrm{Gr}(\ell)/3^\ell$	$r(\ell)$	$\mathrm{Gr}(\ell)/r(\ell)$	$\mathrm{Gr}(\ell)/\mathrm{Gr}(\ell-1)$
3	11	0.4074	6.1	1.7934	–
4	36	0.4444	23.1	1.5557	3.273
5	113	0.4650	75.5	1.4968	3.139
6	346	0.4746	230.8	1.4994	3.062
7	1030	0.4710	681.0	1.5124	2.977
8	2995	0.4565	1967.4	1.5223	2.908
9	8580	0.4359	5605.2	1.5307	2.865
10	24303	0.4116	15816.8	1.5365	2.833
11	68283	0.3855	44324.0	1.5405	2.810
12	190698	0.3588	123572.5	1.5432	2.793
13	530152	0.3325	343156.7	1.5449	2.780
14	1468691	0.3071	949999.5	1.5460	2.770

The root $r(\ell)$. In [2], $h(z) = (-\pi)^{\ell-1} H_\ell(\log z/\pi)\,(\log^2 z + \pi^2)^{-\ell}$ and $r(\ell) = \exp(\pi\rho_\ell)$ with ρ_ℓ the largest root of H_ℓ (they note $\rho_3 = \sqrt{3}/3$). Expanding the explicit expression for h in [2] gives $h(z) = (-1)^{\ell-1} \operatorname{Im}\left[(\log z + i\pi)^\ell\right]/(\pi(\log^2 z + \pi^2)^\ell)$, so its real zeros are the $\log z = \pi \cot(k\pi/\ell)$, $k = 1, \ldots, \ell - 1$, and so $\rho_\ell = \cot(\pi/\ell)$ and $r(\ell) = \exp(\pi \cot(\pi/\ell))$. This is consistent with their table apart from the entry noted above. As $\ell \to \infty$, $\ln r(\ell) = \ell - \pi^2/(3\ell) + O(\ell^{-3})$, so $r(\ell)$ grows like e^ℓ and not 3^ℓ; the upper bound $A(\ell) = \exp\left(\pi\sqrt{(\ell-1)(\ell-2)/6}\right)$ for $r(\ell)$ given in [2] grows like 3.606^ℓ and is far from sharp.

Observations (not theorems). In Table 1, $\mathrm{Gr}(\ell)/3^\ell$ peaks at $\ell = 6$ and then decreases; the ratios $\mathrm{Gr}(\ell)/\mathrm{Gr}(\ell-1)$ decrease from 3.27 at $\ell = 4$ to 2.77 at $\ell = 14$ and $\mathrm{Gr}(\ell)/r(\ell)$ stays between 1.49 and 1.55. This suggests $\mathrm{Gr}(\ell) = \Theta(r(\ell)) = \exp(\ell + O(1))$, i.e. growth rate e and not 3; so the upper bound $0.477 \cdot 3^\ell$ of Conjecture 3.1 should hold for large ℓ while its lower bound fails for all large ℓ. A heuristic stationary-phase computation based on the integral representation of [2] predicts $\mathrm{Gr}(\ell) \sim e^{\ell+1-\gamma}$ (γ Euler's constant); a numerical evaluation of the last sign change of the real-n extension up to $\ell = 100$ gives Gr/r decreasing slowly to about 1.533, to be compared with $e^{1-\gamma} = 1.526$. We have not proved this, and we did not have this evaluation reproduced independently; we include it only as a conjecture.

4 Positivity of the reciprocal of the diagonal Padé denominator of the exponential

4.1 The question

For integers $k \geq 0$, $m \geq 1$ let

$$q_{k,m}(z) = \sum_{j=0}^{m} (-1)^j \binom{m}{j} \frac{(k+m-j)!}{(k+m)!}\, z^j$$

be the denominator of the $[k/m]$ Padé approximant to e^z at 0 (normalised by $q_{k,m}(0) = 1$). Fasi and Higham [3] conjectured the following (their Conjecture 3.2) in connection with a forward-error bound for the matrix exponential.

Conjecture 4.1 (Fasi–Higham [3])**.** Let $k, m \in \mathbb{N}$. If m is odd, then all the coefficients of the series expansion of $q_{k,m}(z)^{-1}$ at 0 are positive.

Scardino [4] proved full positivity for $k = 0$ and *eventual* positivity (positivity of c_n for all large n) whenever the positive real zero of $q_{k,m}$ is the unique zero of minimal modulus, a condition he shows holds for each fixed odd m and all sufficiently large k; he also showed that the conjecture is false in general: with $c_n(k,m)$ the coefficients of $q_{k,m}^{-1}$, he gives exact counterexamples $c_{38}(66,5) < 0$, $c_{37}(44,7) < 0$, $c_{42}(45,9) < 0$. He writes that "the counterexamples are not diagonal, and the diagonal case $k = m$, which is the case relevant in practice, remains open." Here $q_{m,m}$ is a normalised reverse Bessel polynomial. We contribute two things: a computer-assisted proof of the diagonal case for $3 \leq m \leq 361$ (Theorem 4.2), and a numerical description of where positivity first fails for fixed m (Section 4.4). The general diagonal statement remains open.

Theorem 4.2 (computer-assisted)**.** *For every odd m with $3 \leq m \leq 361$, all Taylor coefficients at 0 of $1/q_{m,m}(z)$ are positive. (For $m = 1$, $1/q_{1,1}(z) = \sum_n (z/2)^n$.)*

4.2 The certificate

Fix an odd m and write $q = q_{m,m}$, c_n for the coefficients of $1/q$.

Lemma 4.3. *Suppose q has m distinct zeros $\rho_1, \dots, \rho_m$, exactly one of which, $R = \rho_1$, is real and positive, and $|\rho_i| > R$ for $i \geq 2$. Put $B_i = -1/q'(\rho_i)$, $S = \sum_{i\geq 2} |B_i|$, $\rho_{\min} = \min_{i\geq 2} |\rho_i|$. If $B_1 > 0$ and $N_0 = \max\{0, \lceil \ln(S/B_1)/\ln(\rho_{\min}/R) \rceil\}$, then $c_n > 0$ for all $n \geq N_0$. Consequently $c_n > 0$ for all $n \geq 0$ as soon as $c_0, \dots, c_{N_0-1} > 0$.*

Proof. Since $q(0) = 1$ and q has m simple zeros, partial fractions give $1/q(z) = \sum_i \frac{1}{q'(\rho_i)(z-\rho_i)}$ (there is no polynomial part, the numerator having degree $0 < m$), hence $c_n = \sum_i B_i \rho_i^{-n-1}$ for all $n \geq 0$. For $n \geq N_0$ we have $n + 1 > \ln(S/B_1)/\ln(\rho_{\min}/R)$, i.e. $(\rho_{\min}/R)^{n+1} > S/B_1$, i.e. $S\rho_{\min}^{-n-1} < B_1 R^{-n-1}$. Therefore

$$c_n \;\geq\; B_1 R^{-n-1} - \sum_{i \geq 2} |B_i||\rho_i|^{-n-1} \;\geq\; B_1 R^{-n-1} - S\rho_{\min}^{-n-1} > 0. \qquad \square$$

(In the computation we take N_0 one larger than the displayed value to absorb rounding; this only makes the check stronger.)

4.3 The computation

For each odd m in the range we carried out, in rigorous ball arithmetic (Arb through python-flint [22]):

1. isolation of the m complex zeros of q in disjoint balls; exactly one ball is real and positive, giving R; for each of the other balls a certified lower bound of the modulus exceeding R;
2. the residues $B_i = -1/q'(\rho_i)$ (certified balls), with $B_1 > 0$ certified, and the resulting bound N_0 of Lemma 4.3;
3. positivity of $c_0, \ldots, c_{N_0-1}$ by inverting the power series q in ball arithmetic, with the working precision raised until every one of these coefficients is a ball strictly above 0.

The working precision needed grows with m (from 512 bits for $m \leq 11$ to 8192 bits for $m > 300$), because the ratio $\rho_{\min}/R$ tends to 1: it is 1.094 for $m = 3$, 1.008 for $m = 11$ and about $1 + 1.8 \cdot 10^{-5}$ at $m \approx 240$ (Table 2). The runs for $3 \leq m \leq 361$ were recorded in log files (one line per m) in the repository; every one of the 180 odd values of m in this range appears with the result "proved", and no value is missing or failed. The largest threshold is $N_0 = 372{,}888$ at $m = 361$ (at that point c_n for $n < N_0$ are verified by series inversion with 8192 bits).

In addition, a second, independent exact certification of the first two items in the list was run: for every odd $3 \leq m \leq 361$, $\gcd(Q, Q') = 1$ for the integer polynomial $Q = (2m)!\, q$ (so the zeros are simple), the zeros were isolated by Arb's certified routine for integer polynomials (which also certifies the real ones), exactly one zero is real and positive, and all other zeros have certified modulus larger than R.

What is and is not proved. Theorem 4.2 is a theorem for each listed m individually; the proof consists of Lemma 4.3 together with the certified computations above, which are rigorous in the sense of ball arithmetic but are performed by a program (in the repository, `results/pade_positivity`). It does not give the statement for all m. Moreover the gap $\rho_{\min}/R \to 1$ and the growth of N_0 (the ratio N_0/m^2 is 2.2, 2.6, 2.8, 2.9 for $m = 101, 201, 301, 361$) suggest that this method does not extend to all odd m (this is an expectation, not a proved limitation); a proof for all m will probably need a different argument.

Table 2: Data from the certification runs (selected odd m). R is the positive real zero of $q_{m,m}$, $\rho_{\min}/R$ the gap to the next zero in modulus, N_0 the threshold of Lemma 4.3.

m	R	$\min_{i\geq 2}\lvert\rho_i\rvert/R$	N_0
3	4.64437	1.09446	2
11	15.2447	1.00798	128
101	134.537	1.000102	22 697
201	267.086	1.0000258	103 850
301	–	–	251 082
361	–	–	372 888

4.4 Where positivity first fails

For fixed odd m let $k^*(m)$ be the least k for which a negative coefficient appears. We computed $k^*(m)$ by bisection in k with interval arithmetic (each c_n, $n \leq 5k+300$, certified to have a definite sign), assuming that the set of k for which negative coefficients occur is an interval $[k^*, \infty)$; we verified k^*-1 has no negative coefficient in that range and that $k^*, \dots, k^*+19$ all do, and checked contiguity directly only for $15 \leq m \leq 71$ and $k < 400$. For small m the values agree with exact rational arithmetic: $k^*(5) = 66$ (first negative coefficient $n = 38$), $k^*(7) = 44$ ($n = 37$), $k^*(9) = 42$ ($n = 41$), $k^*(19) = 54$ ($n = 68$); the first two are Scardino's examples, and $(k, m, n) = (42, 9, 41)$ is a smaller counterexample than his $(45, 9, 42)$. For $m = 3$ we found no negative coefficient for any $k < 1500$ (checking $n \leq 4k+200$). For larger m the computation gives

$$k^*(m) - 2m \in \{12, 13, 14\} \qquad (109 \leq m \leq 1009 \text{ sampled}),$$

e.g. $k^*(109) = 230$, $k^*(209) = 430$, $k^*(509) = 1031$, $k^*(1009) = 2032$, and the first negative coefficient occurs near $n \approx k+m-3$. For $m = 109$ and $m = 209$ the thresholds were re-checked with exact rational arithmetic around k^* (no negative coefficient among c_n, $n \leq k+m+10$ for $k = 229, 429$; negative coefficients $c_{337}(230, 109)$ and $c_{637}(430, 209)$). In particular the diagonal $k = m$ is far from the boundary of the failure region for large m ($k^*/m \to 2$ numerically). This law is a numerical observation, not a theorem.

Question 4.4. Is $1/q_{m,m}$ positive for every odd m? Is $k^*(m) = 2m + O(1)$?

5 Series of products of consecutive harmonic numbers

5.1 The question

Let $H_n = 1 + \frac{1}{2} + \cdots + \frac{1}{n}$ and, for $q \geq 1$,

$$S_q = \sum_{k=1}^{\infty} \frac{H_k H_{k+1} \cdots H_{k+q}}{k(k+1)\cdots(k+q)}.$$

Known values (as recalled in Chen and Nguyen [5]) are $S_1 = \zeta(2) + 2\zeta(3)$ and $S_2 = -\frac{1}{2}\zeta(2) + \frac{5}{4}\zeta(3) + \frac{5}{8}\zeta(4)$, and Chen and Nguyen prove

$$S_3 = -\tfrac{4}{9}\zeta(2) - \tfrac{1}{6}\zeta(2)\zeta(3) - \tfrac{7}{24}\zeta(3) + \tfrac{191}{144}\zeta(4) + \tfrac{1}{2}\zeta(5).$$

Furdui and Sîntămărian conjectured that S_q is always a rational linear combination of $\zeta(2), \dots, \zeta(q+2)$ (stated as Conjecture 1 in [5], which we have not read in the original); the value of S_3 shows that this holds for $q = 3$ only if $\zeta(2)\zeta(3) \in \operatorname{span}_{\mathbb{Q}}\{\zeta(2), \dots, \zeta(5)\}$, which is open. Chen and Nguyen end with: "Perhaps we can learn more by computing the closed form for S_q ($q \in \mathbb{N}$)." We do this for $q \le 8$.

5.2 Method

The method is standard in outline (reduction of sums of products of harmonic numbers over rational weights to Euler sums and then to multiple zeta values, in the spirit of Flajolet–Salvy [6] and Xu–Wang [7]); what we contribute is its execution in exact arithmetic for the specific family S_q.

Multiple zeta values are $\zeta(s_1, \dots, s_r) = \sum_{n_1 > \cdots > n_r \ge 1} n_1^{-s_1} \cdots n_r^{-s_r}$ ($s_1 \ge 2$). Fix q and a cut-off K, and let $S_q(K)$ be the sum over $k \le K$.

1. Write $H_{k+j} = H_k + \sum_{i=1}^{j} \frac{1}{k+i}$ ($0 \le j \le q$) and expand the product: $\prod_{j=0}^{q} H_{k+j} = \sum_m H_k^m \, c_m(k)$ with c_m rational functions of k with poles in $\{-1, \dots, -q\}$. Multiplying by $\prod_j (k+j)^{-1}$ and splitting into partial fractions (in exact rational arithmetic) gives a finite $\mathbb{Q}$-linear combination of sums $\sum_{k \le K} H_k^m (k+a)^{-s}$ with integers $0 \le a \le q$ and $s \ge 1$; the individual sums with $s = 1$ diverge as $K \to \infty$ and only the combination converges.

2. A shifted sum is reduced to unshifted ones: with $p = k + a$ one has $H_{p-a} = H_p - \sum_{i=0}^{a-1} \frac{1}{p-i}$, which lowers the power of H by expanding; induction on the power m terminates in sums $\sum_p (p+b)^{-u}$ with explicit rational boundary terms. The changes of the upper limit contribute $O(\log^m K / K)$.

3. The unshifted sums $\sum_{p \le K} H_p^m \, p^{-s}$ are rewritten as truncated nested sums (quasi-shuffle algebra of the truncated multiple zeta values $\zeta_K(s_1, \dots, s_r)$, which satisfy the stuffle relations *exactly* at finite K); those with $s_1 = 1$ are polynomials in $T = H_K$ plus $o(1)$ (the standard stuffle regularisation of multiple zeta values, with $T = H_K$ exactly). Since $S_q(K)$ converges, all positive powers of T in the resulting expression must cancel; our program asserts this cancellation identically in exact rational arithmetic, and it always holds.

4. The output is a $\mathbb{Q}$-linear combination of multiple zeta values. In step 1 a term with H_k^m carries $q + 1 - m$ factors of the form $1/(k+i)$, $1 \le i \le q$, and the factor $\prod_j (k+j)^{-1}$ has simple poles; hence the pole at $k = -a$ has order $s \le q + 2 - m$, and the sums $\sum H_p^m p^{-s}$ that arise have weight $m + s \le q + 2$. The shift reductions of step 2 do not increase the weight.

Proposition 5.1 (computer-assisted)**.** *For $1 \le q \le 9$ the procedure above produces an explicit $\mathbb{Q}$-linear combination of multiple zeta values of weight at most $q + 2$, computed in exact rational arithmetic by a program in the repository (which asserts the cancellation of all divergent terms). For each such q the combination agrees with an independent high-precision numerical evaluation of S_q to about 10^{-62}. Thus the statement "S_q equals this combination" is established by the algorithm, whose correctness rests on the argument above and on the exactness of the arithmetic, and is corroborated numerically; it is not a formal proof checked by a proof assistant.*

The same procedure applies verbatim for every q; we do not claim a theorem for all q because we have only executed it for $q \le 9$.

5.3 Closed forms

Every multiple zeta value of weight ≤ 7 is a polynomial with rational coefficients in single zeta values $\zeta(n)$ (a classical result, available through the standard tables of multiple zeta values), and at weight 8 one new generator, for instance $\zeta(5,3)$, is needed. We found the rational coefficients of these reductions by PSLQ at 150 digits (the MZVs being computed with PARI/GP), so the closed forms below are *numerical identifications* of exact reductions; we label them accordingly. They were then checked against independent numerics (45 digits for $q \leq 8$).

$$S_4 = \tfrac{1}{432}\zeta(2) - \tfrac{45}{64}\zeta(3) + \tfrac{1799}{8640}\zeta(2)^2 - \tfrac{11}{72}\zeta(2)\zeta(3) + \tfrac{13}{18}\zeta(5) + \tfrac{1}{60}\zeta(2)^3 - \tfrac{1}{48}\zeta(3)^2,$$
$$S_5 = \tfrac{83}{16200}\zeta(2) - \tfrac{26657}{86400}\zeta(3) - \tfrac{9811}{64800}\zeta(2)^2 + \tfrac{347}{43200}\zeta(2)\zeta(3) + \tfrac{1669}{2880}\zeta(5) + \tfrac{61633}{756000}\zeta(2)^3 - \tfrac{137}{1440}\zeta(3)^2 - \tfrac{1}{300}\zeta(2)^2\zeta(3) - \tfrac{1}{120}\zeta(2)\zeta(5) + \tfrac{1}{30}\zeta(7).$$

Numerically $S_4 = 0.21289073764960874091117088329\ldots$ (confirmed to 40 digits by an independent implementation). The values of S_6, S_7, S_8 are listed in the repository (`results/harmonic_products/S_tabl`); the constant $\zeta(5,3)$ first appears in S_7 (the top weight of S_6 is 8 but its weight-8 part reduces to single zeta values). We did not find S_q for $q \geq 4$ in the literature, but as noted the method is standard and these values should be viewed as the explicit evaluation of a known type of sum.

5.4 An observation on the top-weight part

In the exact MZV expressions produced for $q \leq 9$, the terms of weight $q+2$ add up to

$$\frac{1}{q!}\sum_{n\geq 1}\frac{H_n}{n^{q+1}} = \frac{\zeta(q+2)+\zeta(q+1,1)}{q!} = \frac{1}{q!}\Big[\frac{q+3}{2}\zeta(q+2) - \frac{1}{2}\sum_{\substack{a+b=q+2\\ a,b\geq 2}}\zeta(a)\zeta(b)\Big].$$

(The last equality is Euler's formula for $\sum H_n/n^s$.) We verified this identity exactly for $q \leq 9$; a heuristic explanation (only partial fractions with all extra harmonic terms placed on a single pole preserve the weight, and a binomial identity collapses them) is not a proof, so we state it as an observation and leave the general case open (Section 9). It shows, in particular, that for $q = 3$ the top-weight part contains $\zeta(2)\zeta(3)$, as in [5].

6 Three questions of Finch on iteration

Finch has recently posed several questions about iterated maps [8, 9, 10]. We answer three of them. The first is an elementary consequence of classical asymptotics of Fatou coordinates and is presented as such; the other two are an exact series with a numerical evaluation and a table of numerical values.

6.1 The constant δ between two solutions of Abel's equation

Setting. Let $\theta(x) = x + \sum_{m\geq 1} c_m x^{m\tau+1}$ be analytic near 0, with an integer $\tau \geq 1$ and $c_1 = \gamma < 0$, so that 0 is an attracting parabolic fixed point. Put $\kappa = -\tau\gamma > 0$ and

$$a = \frac{\tau+1}{2} - \frac{c_2}{\gamma^2}.$$

Let $x_0 > 0$ be a starting point in the basin, $x_n = \theta^n(x_0)$, so $x_n \downarrow 0$. Finch considers two numerical solutions of Abel's equation $g(\theta(x)) = g(x) + 1$.

- The *EJ solution* (Écalle–Jagy). There is a unique formal series $G(x) = \kappa^{-1}x^{-\tau} + a\ln x + \sum_{m\ge 1} b_m x^{m\tau}$ (no constant term) with $G(\theta(x)) = G(x)+1$ (see the proof below). One sets $g_{\mathrm{EJ}}(x_0) = \lim_{n\to\infty}\left[G_K(x_n) - n\right]$ where G_K is the truncation of G after the term $b_K x^{K\tau}$ ($K \ge 1$; the limit exists and does not depend on K, see below). This is Finch's description $g(x) = \lim[g(x_n) - n]$.

- The *ML solution* (Mavecha–Laohakosol), given by the limit

$$\tilde g(x_0) = -\tau \lim_{n\to\infty} n^{1+1/\tau}\Big[\kappa^{1/\tau}x_n - n^{-1/\tau} + \frac{a}{\tau^2}\,\frac{\ln n}{n^{1+1/\tau}}\Big].$$

  Finch displays this formula for the maps he studies, e.g. $\tilde g_{13}(x) = -2\lim n^{3/2}(\sqrt{2}x_n - n^{-1/2} + \frac{3}{8}\ln n\, n^{-3/2})$ for $\theta_{13}(x) = x(1-x^2)$ (here $\tau = 2$, $\kappa = 2$, $a = \frac{3}{2}$), and the corresponding formulas for $x(1-x^k)$ and for $x(1-x)(1+\rho x)$; in all of these cases it agrees with the formula above. We take the formula above as the definition of $\tilde g$ for general θ.

He conjectured exact values of $\delta = g_{\mathrm{EJ}} - \tilde g$ in many cases, in particular $\delta_{13} = -\frac{3}{4}\ln 2$, $\delta_{16} = -\frac{2}{3}\ln 3$, $\delta_{17} = -\frac{5}{9}\ln 6$, and (Addendum III of [8]) for $\theta(x) = x(1-\sigma x^\tau - \rho x^{2\tau})$, $\sigma > 0$:

$$\delta = \Big(\frac{\tau+1}{2\tau} + \frac{\rho}{\tau\sigma^2}\Big)\ln\frac{1}{\tau\sigma},$$

remarking that "δ seems to be completely determined by the parameter τ and two coefficients c_1 & c_2". All these are consequences of the following statement.

Proposition 6.1. *In the setting above, the formal series G exists and is unique, the limits defining $g_{\mathrm{EJ}}(x_0)$ and $\tilde g(x_0)$ exist, and*

$$\delta = g_{\mathrm{EJ}}(x_0) - \tilde g(x_0) = -\frac{a}{\tau}\ln\kappa = \Big(\frac{\tau+1}{2\tau} - \frac{c_2}{\tau\gamma^2}\Big)\ln\frac{1}{\tau|\gamma|} \qquad \textit{for every } x_0.$$

For $\theta = x(1-\sigma x^\tau - \rho x^{2\tau})$ ($c_1 = -\sigma$, $c_2 = -\rho$) this is exactly Finch's Addendum III formula.

Remark 6.2. The logarithmic coefficient $a = \frac{\tau+1}{2} - c_2/c_1^2$ is classical: it is (up to normalisation) Écalle's *résidu itératif* of the parabolic germ, which is the coefficient of the logarithm in the Fatou coordinate (see for example Milnor's survey and Lanford–Yampolsky [12]). Given this, Proposition 6.1 is a short corollary of the definitions of the two algorithms, and we do not claim the asymptotics as new; what we supply is a complete derivation of Finch's conjectured constants.

Proof. The formal series. Put $\varepsilon = x^\tau$. Then

$$\theta(x)^{-\tau} = x^{-\tau}\big(1 + \gamma\varepsilon + c_2\varepsilon^2 + O(\varepsilon^3)\big)^{-\tau} = \varepsilon^{-1} - \tau\gamma + \Big(\tfrac{\tau(\tau+1)}{2}\gamma^2 - \tau c_2\Big)\varepsilon + O(\varepsilon^2),$$

$$\ln\frac{\theta(x)}{x} = \gamma\varepsilon + O(\varepsilon^2), \qquad \theta(x)^{m\tau} - x^{m\tau} = m\tau\gamma\,\varepsilon^{m+1} + O(\varepsilon^{m+2}).$$

For $G = Ax^{-\tau} + a' \ln x + \sum_{m\ge1} b_m x^{m\tau}$ we get $G(\theta(x)) - G(x) = -A\tau\gamma + \big[A(\frac{\tau(\tau+1)}{2}\gamma^2 - \tau c_2) + a'\gamma\big]\varepsilon + \sum_{j\ge2}(\cdots)\varepsilon^j$, where the coefficient of ε^j for $j \ge 2$ contains b_{j-1} with the nonzero factor $(j-1)\tau\gamma$ plus terms involving only $A, a', b_1, \dots, b_{j-2}$. Equating with 1 gives successively $A = -1/(\tau\gamma) = 1/\kappa$; then $a' = -A(\frac{\tau(\tau+1)}{2}\gamma^2 - \tau c_2)/\gamma = \frac{\tau+1}{2} - \frac{c_2}{\gamma^2} = a$ (using $A = -1/(\tau\gamma)$); then $b_1, b_2, \dots$ uniquely. So G exists, is unique, and the truncation satisfies $G_K(\theta(x)) - G_K(x) - 1 = O(x^{(K+1)\tau})$.

Existence of g_{EJ}. Since $x_n \to 0$ monotonically and the recurrence gives $x_{n+1}/x_n = 1 + \gamma x_n^\tau + O(x_n^{2\tau})$, we have $x_n^{-\tau} = \kappa n(1 + o(1))$ (indeed $x_{n+1}^{-\tau} - x_n^{-\tau} = \kappa + O(x_n^\tau) \to \kappa$, and Cesàro's lemma applies), so $x_j^{(K+1)\tau} = O(j^{-(K+1)})$ is summable for $K \ge 1$. Hence $G_K(x_n) - n - G_K(x_0) = \sum_{j<n}[G_K(x_{j+1}) - G_K(x_j) - 1]$ converges, so $g_{\mathrm{EJ}}(x_0)$ exists; it does not depend on $K \ge 1$ because $b_{K+1} x_n^{(K+1)\tau} \to 0$. Moreover, since $b_m x_n^{m\tau} \to 0$,

$$\kappa^{-1} x_n^{-\tau} + a \ln x_n = n + g_{\mathrm{EJ}}(x_0) + o(1). \tag{1}$$

Asymptotics of the orbit. From (1), $x_n^{-\tau} = \kappa n + \kappa g_{\mathrm{EJ}} - \kappa a \ln x_n + o(1)$. Since $x_n \to 0$ and $\ln x_n = O(\ln x_n^{-\tau})$ we first get $x_n^{-\tau} = \kappa n + O(\ln n)$, hence $\ln x_n = -\frac{1}{\tau}\ln(\kappa n) + o(1)$, and then

$$x_n^{-\tau} = \kappa n\,(1 + \varepsilon_n), \qquad \varepsilon_n = \frac{\frac{a}{\tau}\ln(\kappa n) + g_{\mathrm{EJ}}(x_0)}{n} + o\Big(\frac{1}{n}\Big) = O\Big(\frac{\ln n}{n}\Big).$$

Therefore $\kappa^{1/\tau} x_n = n^{-1/\tau}(1 + \varepsilon_n)^{-1/\tau} = n^{-1/\tau}\big[1 - \frac{1}{\tau}\varepsilon_n + O(\ln^2 n/n^2)\big]$ and

$$\kappa^{1/\tau} x_n - n^{-1/\tau} + \frac{a}{\tau^2}\frac{\ln n}{n^{1+1/\tau}} = -\frac{1}{\tau}\, n^{-1-1/\tau}\Big[\frac{a}{\tau}\ln\kappa + g_{\mathrm{EJ}}(x_0)\Big] + o(n^{-1-1/\tau}),$$

where the $\ln n$ terms cancelled. Multiplying by $-\tau n^{1+1/\tau}$ shows that the limit defining $\tilde g(x_0)$ exists and equals $g_{\mathrm{EJ}}(x_0) + \frac{a}{\tau}\ln\kappa$. □

Remark 6.3 (Checks). The proposition reproduces every value in Finch's paper that we tested: δ_{13}, $\delta_{14} = 0$, δ_{15}, δ_{16}, δ_{17}, $\delta(k) = -\frac{k+1}{2k}\ln k$ for $x(1-x^k)$ and for $x(1-x)^k$, $\delta(\rho)$, and the transcendental examples $\ln(1+x)$ $(-\frac{1}{3}\ln 2)$, xe^{-x} (0), $x/(1+x^2)$ $(-\frac{1}{4}\ln 2)$, $\sin x$ $(\frac{3}{5}\ln 3)$ and the Fresnel cosine $(\frac{55}{108}\ln\frac{10}{\pi^2})$. Numerically, Finch's printed 100-digit values $g_{13}(\frac{1}{2}) = 1.18935\ldots$ and $\tilde g_{13}(\frac{1}{2}) = 1.70921\ldots$ differ by $-0.5198604\ldots = -\frac{3}{4}\ln 2$; and a fresh implementation at 40 digits with $4 \cdot 10^5$ orbit steps reproduces -0.5198604 to seven digits (the ML limit converges like $\ln^2 n/n$ and needs extrapolation). These checks use his definitions of the algorithms; we have used his version v6, as an earlier draft (v3) contained a programming error acknowledged in his paper.

6.2 The second derivative $C''(2)$ of an orbit limit

Finch [9] studies $x_{n+1} = x_n + 1 + 1/x_n^2$ and the limit $C(x_0) = \lim_{n\to\infty}(x_n - n)$. He obtains $C(2) = 2.59878685582487\ldots$ from the asymptotic expansion of x_n and, by finite differences of this computation, $C'(2) \approx 0.66156132404868\ldots$ and $C''(2) \approx 0.37462642198301734111$; he notes that an earlier series estimate $0.3909\ldots$ disagrees, and writes: "An independent confirmation of the 0.3746 estimate remains open."

Let x_k be the orbit of x_0, $f(x) = x + 1 + x^{-2}$ so $f'(x) = 1 - 2x^{-3}$. Since $x_k - k \to C$ and $\prod_k f'(x_k)$ converges ($x_k \sim k$, $\sum x_k^{-3} < \infty$), $C'(x_0) = \prod_{k\ge0}(1 - 2x_k^{-3})$. Writing $D_k =$

$\prod_{j<k}(1-2x_j^{-3}) = \mathrm{d}x_k/\mathrm{d}x_0$ and differentiating once more, $\mathrm{d}^2x_k/\mathrm{d}x_0^2 =: E_k$ satisfies $E_{k+1} = f'(x_k)E_k + f''(x_k)D_k^2$, with $f''(x) = 6x^{-4}$, and $C''(x_0) = \lim_k E_k$. Solving the linear recursion,

$$C''(x_0) = C'(x_0)\sum_{j\ge 0}\frac{6\,D_j}{x_j^4\,(1-2x_j^{-3})} = C'(x_0)\sum_{j\ge 0}\frac{6\,D_j}{x_j(x_j^3-2)}. \tag{2}$$

(Finch's series in [9], $C'(x_0)\sum_n (1-2x_n^{-3})^{-1}6x_n^{-4}$ under a uniformity assumption, lacks the factor D_j; this is why it gave 0.3909.) The interchange of the limit $k\to\infty$ with differentiation is justified by the uniform summability of the terms; we have not written out the details, and (2) converges only like $O(1/N)$ in the number N of terms. For $x_0 = 2$ we obtained, with the asymptotic expansion $x_n - n = C - \frac{1}{n} + \frac{C-1/2}{n^2} + \cdots$ (all logarithmic terms vanish) and exact orbit recursions for the derivatives,

$$C''(2) = 0.37462642198301734111 6776\ldots,$$

in agreement with Finch's 0.37462642198301734111 to all 20 digits he printed. A second, independent implementation (fresh code; $N = 10^4$ and $3\cdot 10^4$ orbit steps agree to 33–35 digits) gives 0.37462642198301734111 6776248793, agreeing with the digits above, and a finite-difference evaluation at high precision agrees to 22 digits; the series (2) summed over $2\cdot 10^5$ terms gives 0.3746264219924, consistent with the expected tail $O(1/N)$. None of these computations is interval-certified.

6.3 Half-iterates of b^x for $b = 4/3,\ 5/4,\ 6/5$

Finch [10] writes: "It would be good to see the values of $T(4/3, 1/2)$, $T(5/4, 1/2)$ and $T(6/5, 1/2)$ someday." Here $T(b, x)$ is tetration, and $T(b, \frac{1}{2}) = f^{[1/2]}(1)$ for $f(x) = b^x$, where for $b < e^{1/e}$ we use the regular (Koenigs–Schröder) iteration at the attracting fixed point L ($b^L = L$, $L < e$): with multiplier $\lambda = f'(L) = L\ln b \in (0,1)$ and Koenigs function $\phi(f(x)) = \lambda\phi(x)$, $\phi'(L) = 1$, one puts $f^{[1/2]} = \phi^{-1}(\sqrt{\lambda}\,\phi(\cdot))$. This is the real-analytic half-iterate that is analytic at L; other solutions of $g\circ g = f$ exist and are not analytic at L.

Table 3: Half-iterates $T(b, \frac{1}{2}) = f^{[1/2]}(1)$, $f(x) = b^x$ (regular iteration at the attracting fixed point L). Two independent methods (power series of the inverse Koenigs function; limit formulas at 500-digit working precision) agree to the digits shown.

b	L	λ	$T(b, \frac{1}{2})$
4/3	1.57167727089176302	0.45214337	1.20583743404309559494183930650711256690 0
5/4	1.35220351364068086	0.30173549	1.16359536196500884902594327335843631288 3
6/5	1.25773454137652642	0.22931212	1.13626248672712808418530091864742602558 9

As a check of the convention, the same construction for $b = \sqrt{2}$ (attracting fixed point $L = 2$, $\lambda = \ln 2$) reproduces the value $f^{[1/2]}(1) = 1.24362162766852180429509898\ldots$ printed in [10], Section 1.5 (in a quick check with 120 iterations, to about 20 digits). The values are numerical, not interval-certified: the two methods agree to at least 50 digits, the functional equation $g(g(x)) = b^x$ holds to better than 10^{-85} at several points, and the digits shown are well inside the observed convergence.

7 Gurland's ratio of gamma functions

Wiśniewska [13] studies the modified Gurland ratio

$$G^\star(x,y)=\frac{\Gamma(1+x)\,\Gamma(1+y)}{\Gamma^2\!\left(1+\frac{x+y}{2}\right)},\qquad x,y>0,$$

and proves, for integers $m\ge 2$, the finite expansion

$$\ln G^\star(x,y)=S_m(x,y)+R_{nm}(x,y),\qquad S_m(x,y)=\sum_{k=1}^{m-1}\frac{1}{k}\Big(\frac{x-y}{2}\Big)^{2k}\zeta\Big(2k,1+\frac{x+y}{2}\Big),$$

with $0\le R_{nm}(x,y)\le \varepsilon_n^m(x,y)$, where $\zeta(s,a)=\sum_{n\ge0}(n+a)^{-s}$ is the Hurwitz zeta function. From this she deduces $\ln G^\star=S_\infty:=\lim_m S_m$ under the condition $|x-y|<2(1+\sqrt{xy})$ and ends Section 2 with three open problems: (1) "Does $R_{nm}(x,y)\to 0$ hold for every $x,y>0$?" (2) "Does $\ln G^\star(x,y)=S_\infty(x,y)$ hold generally, and if not, how large the difference can be?" (3) "How is $t(x,y)$ located in the interval $[\sqrt{xy},\frac{x+y}{2}]$?" where t is defined by $\ln G^\star=\big(\frac{x-y}{2}\big)^2\zeta(2,1+t)$ (her Corollary 3).

7.1 Problems 1 and 2

Proposition 7.1. *For all $x,y>0$, with $h=\frac{x+y}{2}$, $d=\frac{x-y}{2}$,*

$$\ln G^\star(x,y)=\sum_{k=1}^{\infty}\frac{1}{k}\,d^{2k}\,\zeta(2k,1+h),$$

the series converging absolutely, with radius of convergence exactly $1+h$ in the variable d; moreover $|d|\le h+1-(1+\min(x,y))$ for all $x,y>0$, so the convergence is geometric with ratio at most $(d/(1+h))^2<1$. In particular $R_{nm}(x,y)\to 0$ as $m\to\infty$ and $\ln G^\star=S_\infty$ for every $x,y>0$.

Proof. Put $a=1+h$ and $f(z)=\ln\Gamma(a+z)$ for $|z|<a$. Here $\ln\Gamma$ is the branch that is real on $(0,\infty)$; it is holomorphic on the half-plane $\operatorname{Re} w>0$ (the zeros of $1/\Gamma$ are absent and the poles of Γ lie on the non-positive real axis). For $|z|<a$ we have $\operatorname{Re}(a+z)>0$, so f is holomorphic in the disc $|z|<a$, and its Taylor series at 0 converges to f there. Its coefficients are $f^{(j)}(0)=\psi^{(j-1)}(a)$ for $j\ge1$, and $\psi^{(n)}(a)=(-1)^{n+1}n!\,\zeta(n+1,a)$ for $n\ge1$; hence $f^{(2k)}(0)=(2k-1)!\,\zeta(2k,a)$ for $k\ge1$. Since $1+x=a+d$ and $1+y=a-d$, we have $\ln G^\star=f(d)+f(-d)-2f(0)$, and $|d|<a$ holds for all $x,y>0$ because $a-|d|=1+\min(x,y)>1$. Adding the Taylor series at d and $-d$, the odd terms cancel:

$$\ln G^\star=2\sum_{k\ge1}\frac{f^{(2k)}(0)}{(2k)!}d^{2k}=\sum_{k\ge1}\frac{1}{k}\,\zeta(2k,a)\,d^{2k}.$$

The terms are nonnegative, and $\zeta(2k,a)\ge a^{-2k}$, so for $|d|=a$ the series would diverge (it is bounded below by $\sum\frac{1}{k}$); thus the radius is exactly a. For $|d|<a$, $\zeta(2k,a)=a^{-2k}(1+O((a/(a+1))^{2k}))$ and the terms are $O\big(k^{-1}(d/a)^{2k}\big)$. Finally S_m is the $(m-1)$-st partial sum, so $R_{nm}=\ln G^\star-S_m\to0$. □

Thus the answer to Problems 1 and 2 is "yes" and the difference $\ln G^\star - S_\infty$ is 0. The argument is elementary (it is just the radius of convergence of the Taylor series of $\ln\Gamma$ at $1+h$), and we do not regard it as deep; it was not stated in [13], whose sufficient condition $|x-y| < 2(1+\sqrt{xy})$ is therefore not necessary. As a sanity check, for $(x,y) = (0.01, 100)$, where $|d|/a = 0.98$, the series evaluated with 60–120 working digits agrees with $\ln G^\star$ and the remainders behave like the predicted $\sim r^{2N}/(N(1-r^2))$, $r = |d|/a$ (about 10^{-10} after 500 terms).

7.2 Problem 3: partial information (numerical)

Write $g = \sqrt{xy}$, so $g \le t \le h$. From Proposition 7.1, as $d \to 0$ with h fixed,

$$\frac{\ln G^\star}{d^2} = \psi'(1+h) + \frac{d^2}{12}\psi'''(1+h) + O(d^4), \qquad \text{so} \qquad t = h + \frac{d^2}{12}\frac{\psi'''(1+h)}{\psi''(1+h)} + O(d^4),$$

while $h - g = d^2/(2h) + O(d^4)$. Hence the relative position $\vartheta = (h-t)/(h-g)$ of t between h and g satisfies $\vartheta \to \frac{h}{6}\frac{\psi'''(1+h)}{-\psi''(1+h)}$ as $d \to 0$, which tends to $\frac{1}{3}$ as $h \to \infty$ (using $\psi''(z) \sim -z^{-2}$, $\psi'''(z) \sim 2z^{-3}$). So no bound of the form $t \ge h - c\,(h-g)$ can hold with $c < \frac{1}{3}$. In a numerical search (20 000 random points with $x, y \in [10^{-3}, 10^{3.5}]$, the largest 200 values re-evaluated with 40-digit arithmetic) we found $\vartheta < \frac{1}{3}$ with maximum 0.33327; a second search with new code (6 000 random points, 40-digit arithmetic) again found no violation, maximum 0.333277, and for $(x,y) = (10^3, 1010)$, $(10^4, 10010)$, $(10^5, 100010)$ it gave $\vartheta = 0.333167,\ 0.333317,\ 0.333332$, approaching $\frac{1}{3}$ from below. This suggests:

Conjecture 7.2. For all $x \ne y$ in $(0,\infty)$, $t(x,y) > \dfrac{x + y + \sqrt{xy}}{3}$ (equivalently $\vartheta < \frac{1}{3}$). The constant $\frac{1}{3}$ cannot be improved.

We have no proof of this inequality. (It is a statement about a Hurwitz-zeta quantity for which a termwise comparison in the Laplace variable fails.)

8 Completely monotonic degrees of Stirling remainders

8.1 The question

A function $f \in C^\infty(0,\infty)$ is completely monotonic (CM) if $(-1)^j f^{(j)} \ge 0$ for all $j \ge 0$. For $f \to 0$ at infinity, the *real completely monotonic degree* is $d_{CM}(f) = \sup\{\rho \ge 0 : x^\rho f(x) \text{ is CM}\}$. For $n \ge 0$ let

$$R_n(x) = (-1)^n\Big[\ln\Gamma(x) - (x-\tfrac12)\ln x + x - \tfrac12\ln 2\pi - \sum_{k=1}^{n}\frac{B_{2k}}{2k(2k-1)x^{2k-1}}\Big], \qquad x > 0,$$

the signed Stirling remainders, with Bernoulli numbers B_j. Krasniqi [14] proves

$$\tfrac{8}{3} < d_{CM}(R_2) < 3, \qquad \tfrac{32}{9} < d_{CM}(-R_2') < 4$$

(his Theorem 1.2), and, for $\delta_n := d_{CM}(R_n) - (2n-2)$, that $\limsup_n (2n-1)\sup_m \delta_{n,m} \le 2\pi^2/J$ with $J = \int_0^{2\pi}\frac{1-\cos u}{u}\,\mathrm{d}u$ (Theorem 1.3; here $\delta_{n,m}$ is the analogous excess for the m-th derivative, and $\delta_{n,0} = \delta_n$). He adds that "the exact real degrees and the optimal asymptotic constants remain open." We computed the degrees numerically. The results below are numerical, not proved.

8.2 Method

For $n \geq 2$, $R_n(x) = \int_0^\infty e^{-xt} h_n(t)\,\mathrm{d}t$ with $h_n(t) = (-1)^n t^{-1}\big[\frac{1}{e^t-1} - \frac{1}{t} + \frac{1}{2} - \sum_{k\leq n} \frac{B_{2k}t^{2k-1}}{(2k)!}\big] \geq 0$, and $-R_n'$ has density $t\,h_n(t)$. By Bernstein's theorem and uniqueness of the Laplace transform, $x^\rho f$ is CM if and only if the Riemann–Liouville derivative $D^\rho h$ of the density is nonnegative on $(0,\infty)$ (for ρ below the order of vanishing of h at 0 there are no boundary terms). We computed $\rho_c = \sup\{\rho : D^\rho h \geq 0\}$; at ρ_c the function $D^\rho h$ first touches zero at a double root $t_* \approx 11.47$ (for R_2) and ≈ 8.04 (for $-R_2'$). Two independent implementations were used (one with Chebyshev interpolation and QUADPACK algebraic-weight quadrature, one with Gauss–Legendre quadrature after a change of variables), and the second implementation additionally checked the result *without* the Laplace representation by computing exact derivatives of $x^\rho f$ (Leibniz rule and polygamma functions at 300 digits) up to order 200: all derivatives of the right sign for $\rho = \rho_c - 0.02$ and $\rho_c - 0.003$, and violations for $\rho = \rho_c + 0.02$.

8.3 Results

Proposition 8.1 (numerical)**.** *To the digits shown (agreement of the two independent implementations to at least 12 digits),*

$$d_{CM}(R_2) = 2.8667042552\ldots, \qquad d_{CM}(-R_2') = 3.7984354776\ldots,$$

both inside the intervals proved in [14].

Table 4 gives $d_{CM}(R_n)$ for further n and the normalised excess $(2n-1)\delta_n$. These values come from our first implementation; the entries marked † were also recomputed by the independent implementation.

Table 4: $d_{CM}(R_n)$ and $(2n-1)\delta_n$, $\delta_n = d_{CM}(R_n) - (2n-2)$; and $d_{CM}(-R_n')$ for small n. (†: recomputed independently.)

n	$d_{CM}(R_n)$	$(2n-1)\delta_n$	$d_{CM}(-R_n')$
2	$2.8667042552^\dagger$	2.6001	$3.7984354776^\dagger$
3	$4.7074638766^\dagger$	3.5373	$5.6534285885^\dagger$
4	6.5993536696	4.1955	7.5587377578
5	$8.5213225063^\dagger$	4.6919	$9.4900039902^\dagger$
6	10.4617284606	5.0790	11.4369053276
8	14.3760290864	5.6404	15.3593656569
10	$18.3171745147^\dagger$	6.0263	–
14	26.2415343007	6.5214	–
18	34.1950137509	6.8255	–
22	$42.1635160400^\dagger$	7.0312	–
27	52.1360471128	7.2105	–

Numerical observation on the sharpness of Krasniqi's constant. The numbers $(2n-1)\delta_n$ increase monotonically with n in the table. Extrapolating the sequence for $n = 2,\ldots,27$ by least-squares polynomials of degree 4–6 in $1/n$ or $1/(n+\frac{1}{2})$ (using the data for $n \geq 8, 10, 12$)

gives limits between 8.0967 and 8.0977, median 8.0975, to be compared with Krasniqi's bound $2\pi^2/J = 2\pi^2/\mathrm{Cin}(2\pi) = 8.097627\ldots$. The data are therefore compatible with $\lim_n (2n-1)\delta_n = 2\pi^2/J$, i.e. with his upper bound being attained (at $m = 0$); the extrapolation is not a proof, and the spread above is a rough indication of its uncertainty (no error bound).

Caveats. The computations are quadrature computations with the stability checks described, not interval arithmetic; non-negativity of $D^\rho h$ below ρ_c was checked on grids, and the root ρ_c was located as a tangency (double root) of $D^\rho h$. The entries marked † agree with the independent implementation to at least 10 digits (for $n = 22$ the independent value needed many quadrature nodes to converge; its quoted uncertainty is $3 \cdot 10^{-11}$); the other entries come from one implementation only.

9 Open problems

We collect questions that our work leaves open. Items 1–3 are conjectures we could neither prove nor refute; the others are open questions from the cited papers on which we have only partial information.

1. **Diagonal Padé positivity.** Is $1/q_{m,m}(z)$ positive (all Taylor coefficients) for every odd m? Theorem 4.2 covers $3 \le m \le 361$. Is $k^*(m) = 2m + O(1)$ (Section 4.4)? The method of Section 4 is unlikely to reach all m, since the gap $\rho_{\min}/R$ seems to tend to 1.

2. **Growth of $\mathrm{Gr}(\ell)$.** Prove that $\log \mathrm{Gr}(\ell) = \ell + O(1)$, or the sharper observed law $\mathrm{Gr}(\ell) \sim e^{\ell+1-\gamma}$ (Section 3). In particular decide whether the upper bound $\mathrm{Gr}(\ell) < 0.477 \cdot 3^\ell$ of [2] holds for all $\ell \ge 5$ (it holds for $5 \le \ell \le 14$).

3. **Gurland's inequality.** Is $t(x, y) > (x + y + \sqrt{xy})/3$ for all $x \ne y$ (Conjecture 7.2)? This would answer the third open problem of [13] sharply.

4. **Nonnegativity for the Painlevé-type expansion.** For $u''u - (u')^2 = e^{-x}P(u) - 1$ with $P(0) = 0$, $P'(0) = \frac{1}{2}P''(0)$ and all coefficients of P nonnegative, are all coefficients of the polynomials p_k in $u = \sum_k p_k(x)e^{-kx}$ nonnegative (Conjecture 7.2 of [15]), and does the series converge (Conjecture 7.3 there)? The existence of the polynomials p_k is proved in [16]. Our exact rational computations (not independently reproduced) found nonnegative coefficients in all cases tried: $P = t^3$ up to $k = 40$, $P = t^4$ up to $k = 30$, $P = t^6$ up to $k = 20$.

5. **Sharpness in Krasniqi's asymptotic bound.** Is $\lim_{n\to\infty}(2n-1)\,(d_{CM}(R_n) - (2n-2)) = 2\pi^2/\mathrm{Cin}(2\pi)$ (Section 8)? The numerical evidence is consistent with the bound of [14] being attained.

6. **Harmonic products.** Does the top-weight law weight-$(q+2)$ part of $S_q = \frac{1}{q!}\sum_{n\ge1} H_n/n^{q+1}$ hold for all q (checked exactly for $q \le 9$; Section 5)? Is $\zeta(2)\zeta(3)$ in the $\mathbb{Q}$-span of $\zeta(2), \ldots, \zeta(5)$ (this is equivalent to the conjecture of Furdui and Sîntămărian holding for $q = 3$, see [5])?

7. **Uniqueness in Fan's conjecture.** Is the equilibrium measure of the vertex set of the regular N-gon, N odd, $r > 2$, unique up to rotation? Theorem 2.3 gives existence of an equilibrium measure with the stated support and masses.

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